Chapter 2. The Method of Mathematical Induction

One Step at a Time

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2.1 Weak induction

Motivation. To prove a statement for every natural number it suffices to check the first case and show that each case implies the next. Induction turns one infinite task into two finite ones; its price is that the statement to be proved must already be known or guessed.

The method. To prove a statement $S(m)$ for all natural m:

1. *Initial step:* prove $S(m)$ for the smallest m for which it is defined (usually 0 or 1).
2. *Induction step:* prove that $S(n)$ (the *induction hypothesis*) implies $S(n+1)$.

Validity. If both steps are done but S fails somewhere, let $m+1$ be the smallest failure. By the initial step, $m+1$ is not the first index, so $S(m)$ is defined and true by minimality; the induction step with $n = m$ then gives $S(m+1)$, a contradiction.

Example 2.1

For all positive integers m,

$$1^2 + 2^2 + \cdots + m^2 = \frac{m(m+1)(2m+1)}{6}. \tag{2.1}$$

Solution. For $m = 1$ both sides are 1. Assume (2.1) for n. The statement for $n+1$ differs from it by adding $(n+1)^2$ on the left and

$$\frac{(n+1)(n+2)(2n+3) - n(n+1)(2n+1)}{6} = \frac{(n+1)\big[(n+2)(2n+3) - n(2n+1)\big]}{6} = (n+1)^2$$

on the right. Adding this identity to (2.1) for n gives the statement for $n+1$.

Example 2.2

Let $f(m)$ be the maximum number of regions into which m straight lines divide the plane. Then $f(m) = \frac{m(m+1)}{2} + 1$.
Solution. $f(1) = 2$. Given $n+1$ lines, look at the last line s. It crosses $k \le n$ of the other lines, so it passes through $k+1$ regions formed by the others and cuts each of them in two. Hence $f(n+1) \le f(n)+n+1$, with equality when s meets all other lines in distinct new points (lines in general position). Thus $f(n+1) = f(n)+n+1 = \frac{n(n+1)}{2}+1+n+1 = \frac{(n+1)(n+2)}{2}+1$.

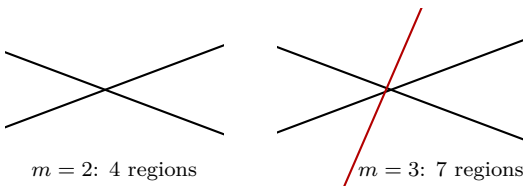


Figure. *Example 2.2: the new (red) line crosses both old lines, passes through 3 old regions and splits each, so* $f(3) = f(2) + 3$.

The formula had to be known in advance; finding it is the weak point of induction. Often it can be guessed from small cases:

Example 2.3

Let $a_0 = 1$ and $a_{n+1} = 3a_n + 1$. Then $a_m = \frac{3^{m+1}-1}{2}$.
Solution. The first values $1, 4, 13, 40, 121$ suggest the formula. It holds for $m = 0$, and if it holds for n, then $a_{n+1} = 3 \cdot \frac{3^{n+1}-1}{2} + 1 = \frac{3^{n+2}-1}{2}$.

A formula like $a_{n+1} = 3a_n + 1$ is a **recurrence relation**; $a_n = \frac{3^{n+1}-1}{2}$ is an **explicit** formula, which is also **closed**: it contains no $\sum$ or $\prod$ over a varying number of terms. For example $f(n) = \sum_{i=1}^{n} i$ is explicit but not closed, while $\frac{n(n+1)}{2}$ is closed. Induction often turns a recurrence into a closed formula.

Sets. A finite **set** is an unordered collection of distinct objects ($\{1, 3, 2\} = \{2, 1, 3\}$); if repetitions are allowed, it is a **multiset**. $B \subseteq A$ means every element of B lies in A. Since enumerative statements usually depend only on the size of a set, we use the canonical n-element set $[n] := \{1, 2, \ldots, n\}$.

Theorem 2.4

For every positive integer n, $[n]$ has 2^n subsets.

Proof. $[1]$ has the two subsets $\emptyset, \{1\}$. Assume the claim for n. Subsets of $[n+1]$ not containing $n+1$ are exactly the subsets of $[n]$: 2^n of them. Those containing $n+1$ are $\{n+1\} \cup T$ with $T \subseteq [n]$ arbitrary: another 2^n. Altogether 2^{n+1}. ■

Pitfalls

The initial step cannot be skipped: "$2n+1$ is even" has a valid induction step ($2(n+1)+1-(2n+1) = 2$) but no true initial case. More subtly, "all horses have the same colour" can be "proved" as follows. Among $n+1$ horses, the first n and the last n are each monochromatic, and they overlap, so all $n+1$ are. But for $n = 1$ the two groups (first horse, last horse) are disjoint, so the step from 1 to 2 fails. The induction step must be valid for *every* n from the initial value on.

2.2 Strong induction

Motivation. Some recurrences express a_{n+1} through all earlier terms, so knowing only a_n is not enough. Strong induction lets the induction step use the statement for *all* smaller values; it is no stronger logically, but it is often the convenient form.

Example 2.5

Let $a_0 = 0$ and $a_{n+1} = a_0 + a_1 + \cdots + a_n + n + 1$ for $n \geq 0$. Then $a_n = 2^n - 1$.
Solution. True for $n = 0$. If $a_k = 2^k - 1$ for all $k \leq n$, then

$$a_{n+1} = \sum_{k=0}^{n} (2^k - 1) + n + 1 = (2^{n+1} - 1) - (n+1) + n + 1 = 2^{n+1} - 1.$$

The method (strong induction).

1. *Initial step:* prove the statement for the smallest n for which it is defined.
2. *Induction step:* prove that the statement for all integers $< n+1$ implies it for $n+1$.

Validity. If $n+1$ were the smallest failure, the statement would hold for all smaller integers (by minimality and the initial step), and the induction step would give it for $n+1$. From now on $\mathbb{N}$ denotes the set of nonnegative integers.

Example 2.6

Let $f(0)=1$, $f(1)=2$ and $f(n+1)=f(n-1)+2f(n)$ for $n \geq 1$. Then $f(n) \leq 3^n$ for all $n \in \mathbb{N}$.
Solution. True for $n=0,1$. If it holds for all values $\leq n$ ($n \geq 1$), then $f(n+1) \leq 3^{n-1}+2\cdot 3^n = 7 \cdot 3^{n-1} < 3^{n+1}$. Both $f(n-1) \leq 3^{n-1}$ and $f(n) \leq 3^n$ are used, so the strong hypothesis is needed.

Variations

It is often convenient to shift the parameter (assume the statement for $n-1$, prove it for n), or to induct over even or odd integers only; these variants are equally valid and are used in later chapters.